YAKEEN

NEET 2.0 2027

Physics | Lecture - 04

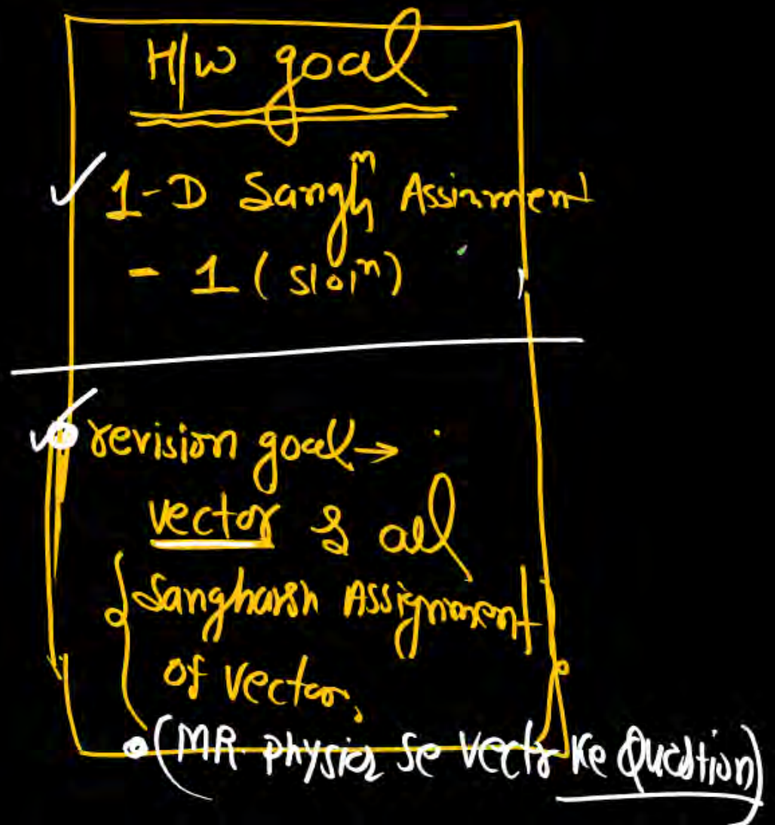
Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays Goal

PhD on speed / velocity



Direction is more important than speed

How fast (no direction known)

दिशा

Velocity (direction / How fast)

Copy at ghar

MR* Box → for graph

Kisi bhi graph se 4 - Bate Pata chalti hai →

#(1) y^{th} - co-ordinate se "y wali P.Q ki value"

#(2) x^{th} - co-ordinate se "x wali P.Q ki value"

#(3) slope se $= \frac{\Delta y}{\Delta x}$ = Jo P.Q defined hoga uska value

#(4) Area se $= \int y\,dx$ = Jo P.Q defined hogo.

MR* Box for U-turn : in 1-D

1-D me U-turn whi hoga Jha velocity zero ho but if V=0 ho wha U-turn ho Must nahi hai ✓

- x-t graph me U-turn wha hoga Jab x → Increasing se decreasing ho Jay or x decreasing se Increasing ho Jay.

My **girlfriend** said she needs some **Time** and **Distance**



Is **she** Calculating **Velocity**??

Speed — How fast

In time 't' — time interval

At time 't' — instantaneous

$$\text{Avg speed} = \frac{\text{Total distance}}{\text{total time}}$$

= scalar

= m/s

⇒ No differention

$$\text{Avg velocity} = \frac{\text{Total displ}^{n}}{\text{total time}}$$

→ vector

$$v = \frac{dx}{dt}$$

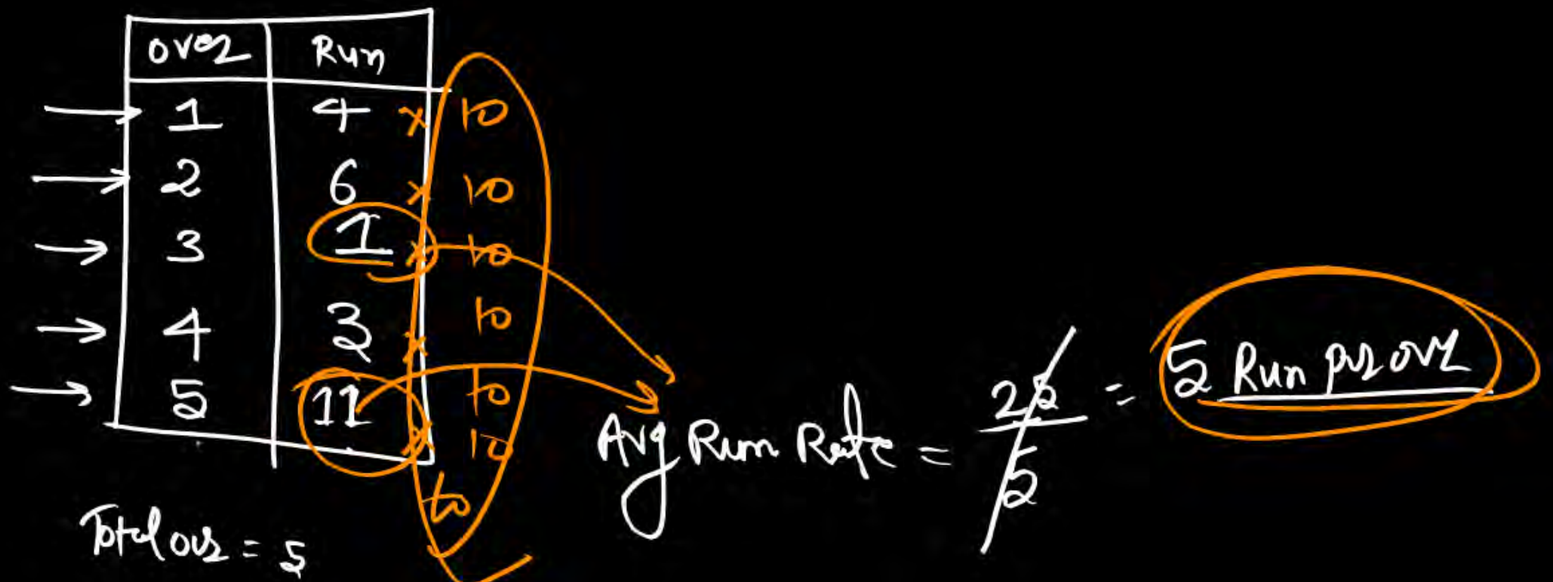
= slope of distance/time graph

= scalar

= m/s.

= No direction

Noccp

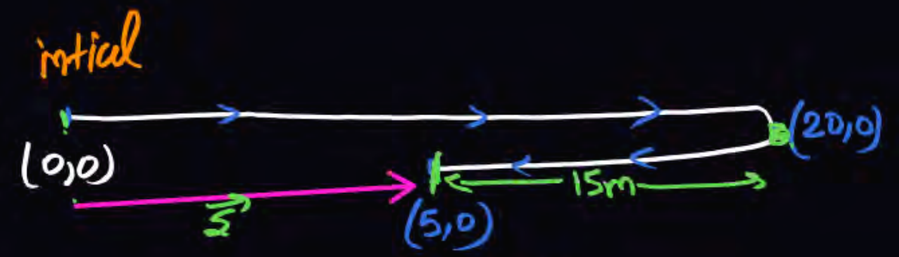

over
Run
1
4
2
6
3
1
4
3
5
11
Total overs = 5
Total Run = 25
Avg Run Rate = 25/5 = 5 Run per over

Find **Avg. speed** and **Avg velocity**

① 

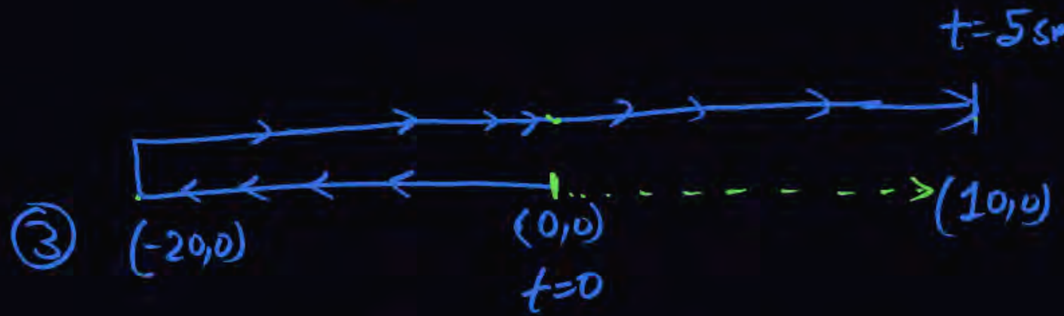
Avg velocity $= \frac{20\hat{i}}{4} = 5\text{m/s}\,\hat{i}$

Avg speed $= \frac{20}{4} = 5\text{ m/s}$.

② 

Avg velocity $= \frac{5-0}{10} = \frac{1}{2}\text{ m/s}$

Avg speed $= \frac{20+15}{10} = \frac{35}{10} = 3.5$

③ 

$\Rightarrow$ |Avg velocity| $= \frac{10}{5} = 2\text{m/s}$

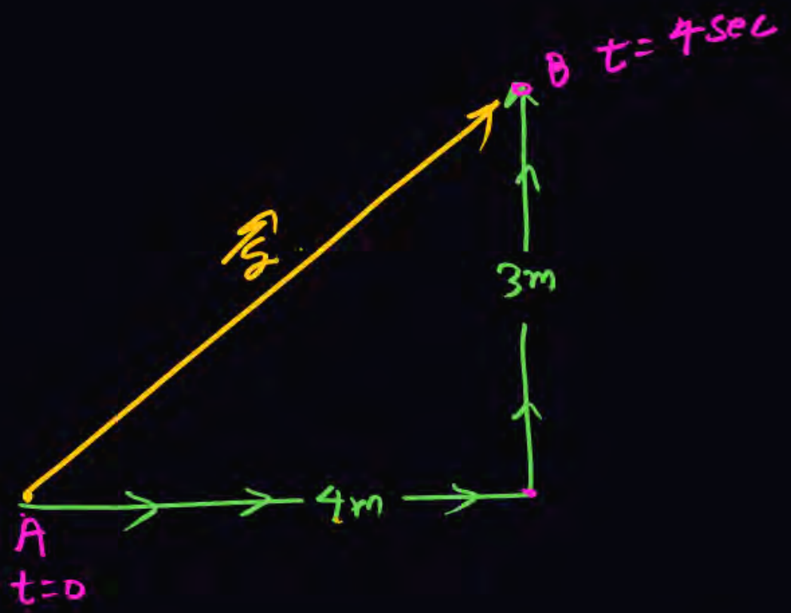
$\Rightarrow$ Avg speed $= \frac{50}{5} = 10\text{m/s}$

⑤ 

Avg speed $= \frac{40}{10} = 4\text{m/s}$

- Avg velocity = zero

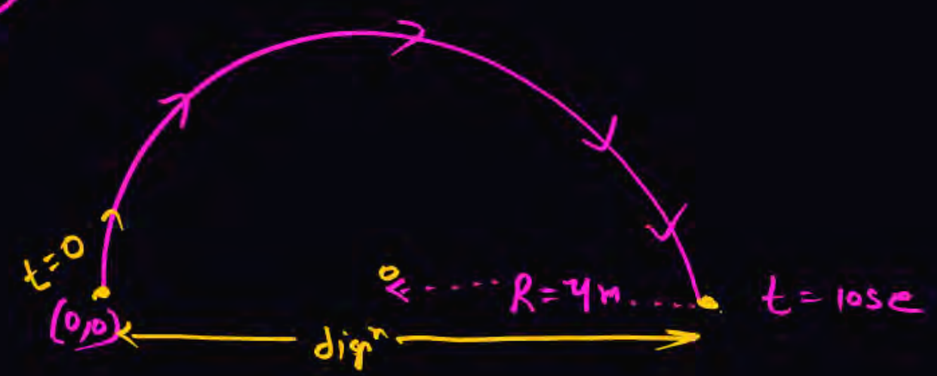
⑤



$$\text{Avg speed} = \frac{7}{4}\ \text{m/s} \checkmark$$

$$|\text{Avg velocity}| = \frac{5}{4}\ \text{m/s}$$

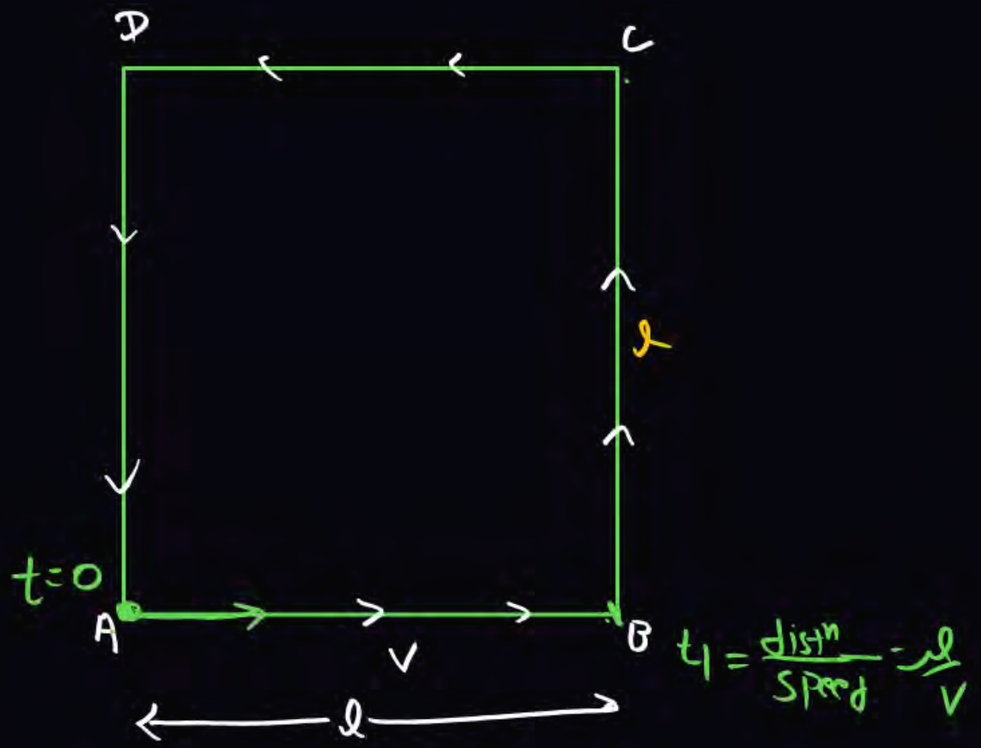
(6)



$$\text{Avg speed} = \frac{\pi R}{10} = \frac{\pi \times 4}{10} \text{ m/s}$$

$$\text{Avg velocity} = \frac{2R}{10} \text{ m/s.}$$

⑦ Object is moving on square park with **Constant speed V** then find Avg. speed & Avg velocity for given path.



$t_1 = \frac{\text{disp}^n}{\text{speed}} = \frac{l}{V}$

time = $\frac{\text{disp}^n}{\text{speed}}$

Motion	disp^n	disp^m ✓	Avg speed (Cost^n Speed = V)	Avg velocity	time
$A \to B$	l	l	V	$\frac{l}{l/V} = V$	$\frac{l}{V}$
$A \to B \to C$	$2l$	$\sqrt{2}l$	V	$\frac{\sqrt{2}l}{2l}V = \frac{V}{\sqrt{2}}$	$\frac{2l}{V}$
$A \to D$	$3l$	l	$\frac{3l}{3l/V} = V$	$\frac{l}{3l}V = \frac{V}{3}$	$\frac{3l}{V}$
$A \to A$	$4l$	0	V	0	$\frac{4l}{V}$

Q

Is magnitude of Avg. velocity always equal to Avg. speed ??

$$|\text{Avg. velocity}| = \text{Avg. speed}\ ??$$

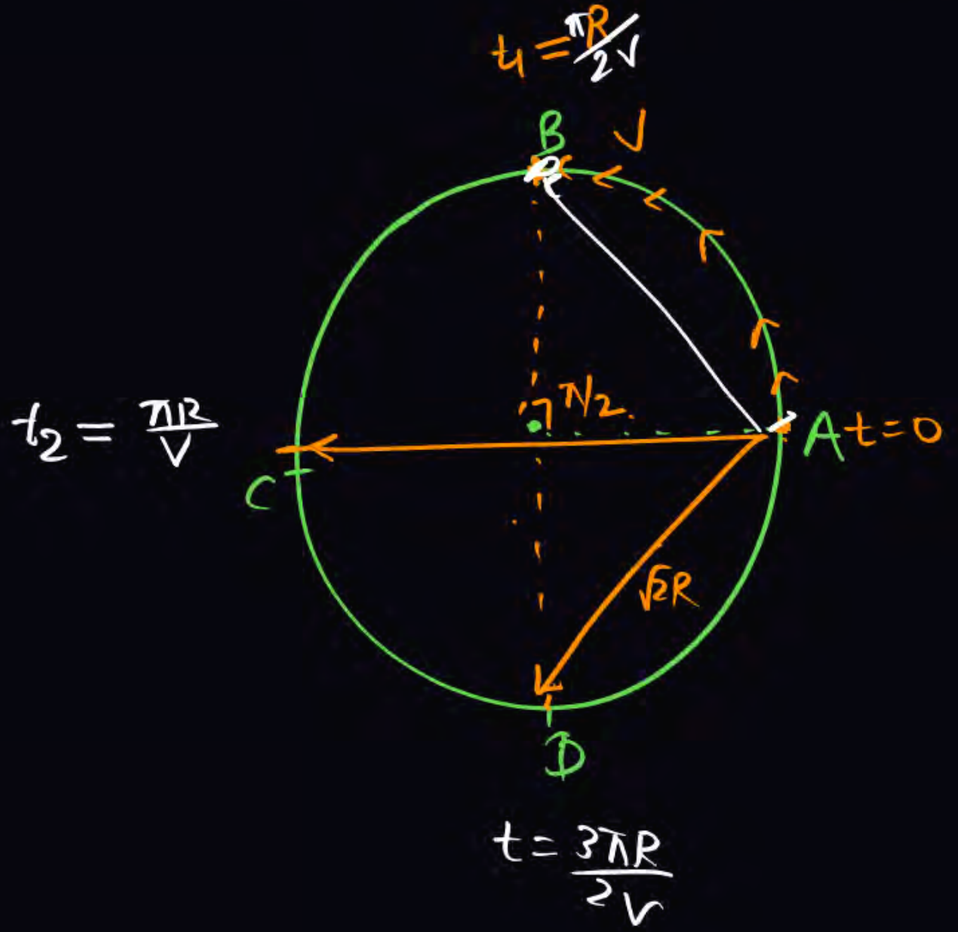
No

only equal when moving on 1-D without change in direction.

$$|\text{Avg velocity}| \leq \text{Avg speed}$$

always valid

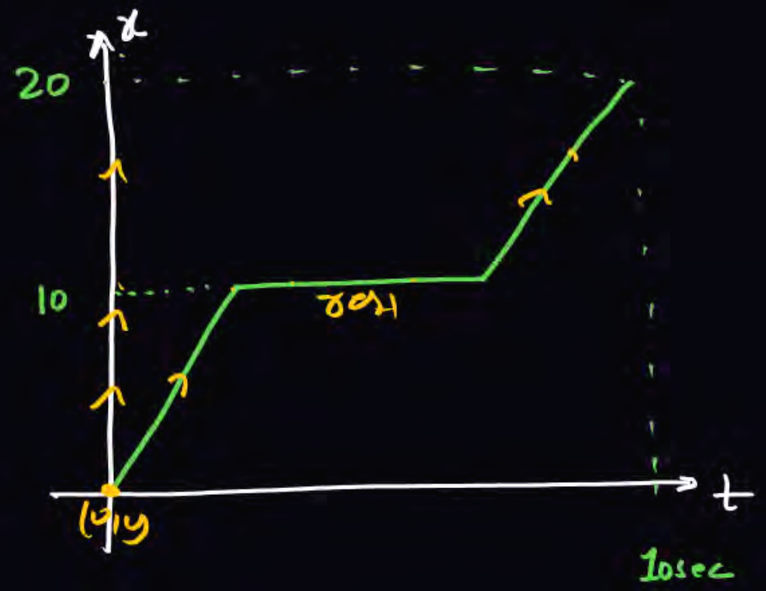
8 Object is moving on circular Path with Constant speed V then find Avg. speed & Avg Velocity for given path.



time = $\frac{\text{disty}}{\text{speed}}$

Motion	disty	dispm	Avg speed	Avg velocity	time
A → B	$\frac{\pi R}{2}$	$\sqrt{2}R$	V	$\frac{\sqrt{2}R\,2V}{\pi R} = \frac{2\sqrt{2}V}{\pi}$	$\frac{\pi R}{2V}$
A → B → C	πR	$2R$	V	$\frac{2RV}{\pi R} = \frac{2V}{\pi}$	$\frac{\pi R}{V}$
A → D	$\frac{3\pi R}{2}$	$\sqrt{2}R$	V	$\frac{\sqrt{2}R\,2V}{3\pi R} = \frac{2\sqrt{2}V}{3\pi}$	$\frac{3\pi R}{2V}$
A → A	$2\pi R$	0	V	0	$\frac{2\pi R}{V}$

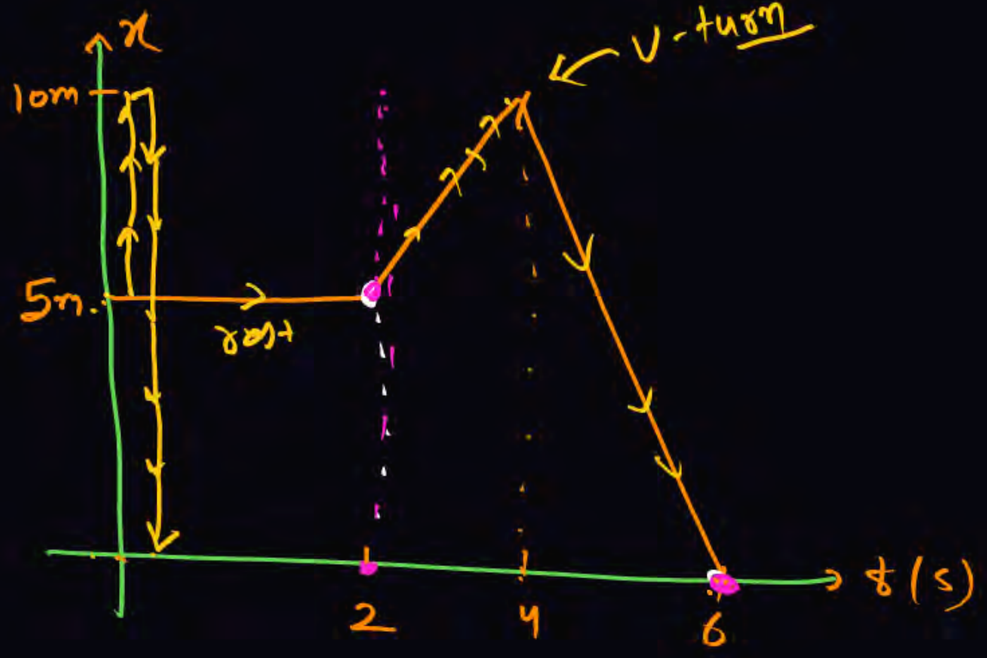
(9)



→ Avg speed in 10 sec $= \frac{20}{10} = 2$ m/s

→ Avg velocity $= \frac{20}{10} = 2$ m/s.

(10)



Avg speed in 6-sec $= \frac{15}{6}$

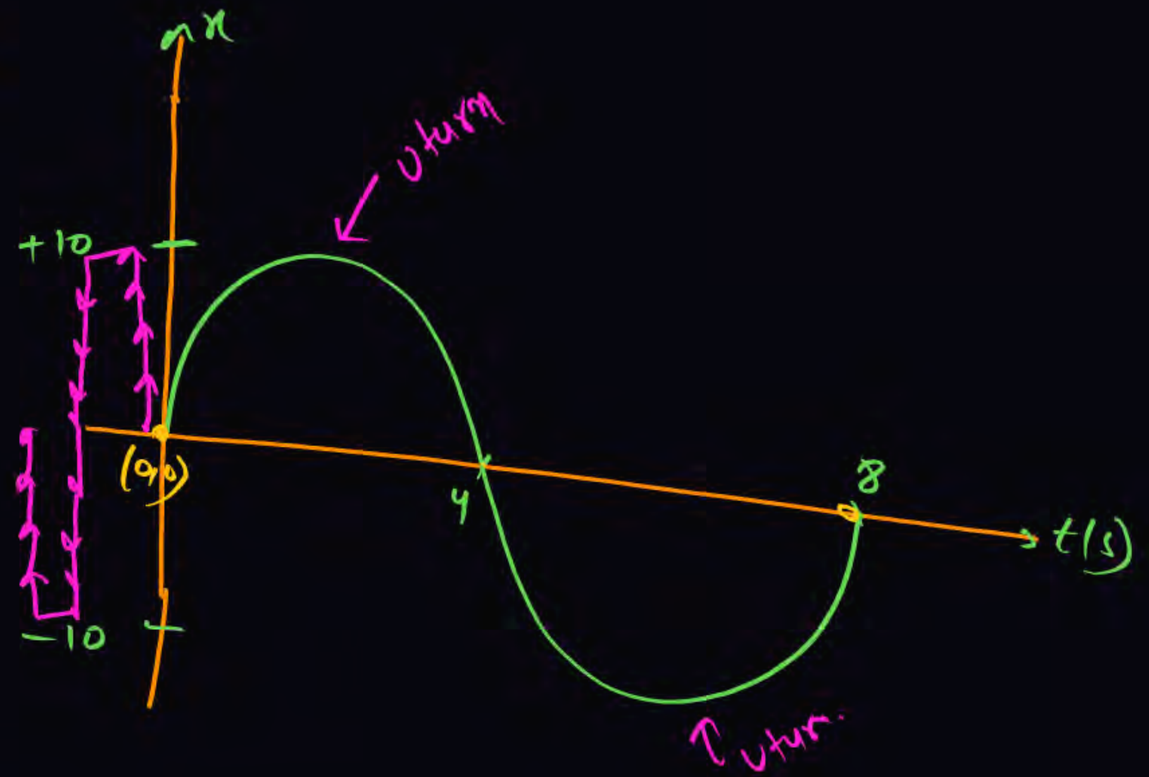
Avg velocity in 6 sec $= \frac{0-5}{6} = -\frac{5}{6}$ m/s

Avg speed B/W 2sec to 6sec $= \frac{15}{4}$ m/s $= \frac{\text{Total dist}^n}{\text{total time}}$

for time (0-4) sec

Avg speed = |Avg velocity|

(11)





(11) $|\text{Avg speed}|_{t=8} = \frac{40}{8} = 5\,m/s$ ✓

~~(11)~~ $|\text{Avg velocity}|_{t=8\,sec} = 0$

(12) Object moves 10m in 2 sec then in same dirn 20m in 3 sec

find Avg speed and velocity.

$$|\text{Avg velocity}| = \text{Avg speed} = \frac{\text{Total distance}}{\text{total time}}$$

$$= \frac{10m + 20m}{2s + 3s}$$

$$= \frac{30}{5} = 6 m/s \checkmark$$

(13) Object moves with speed s_1 for time t_1 then speed s_2 for time t_2 then Avg speed.



$$\text{Avg speed} = \frac{\text{total distn}}{\text{total time}} = \frac{d_1 + d_2}{t_1 + t_2}$$

$$= \frac{s_1 t_1 + s_2 t_2}{t_1 + t_2} \checkmark$$

If equal time intval $t_1 = t_2 = t$

$$\left[\text{Avg speed} = \frac{(s_1 + s_2)\cancel{t}}{2\cancel{t}} = \frac{s_1 + s_2}{2}\right]$$

3- equal time intraval

$$\text{Avg speed} = \frac{s_1 + s_2 + s_3}{3}\Bigg] \checkmark$$

(14) object moves with speed s_1 for distance x_1 then speed s_2 for distance x_2 then Avg speed.



$$\text{Avg speed} = \frac{x_1 + x_2}{t_1 + t_2}$$

$$\boxed{\text{Avg speed} = \frac{x_1 + x_2}{\frac{x_1}{s_1} + \frac{x_2}{s_2}}}$$

→ if equal distance interval $(x_1 = x_2 = x)$

$$\text{Avg speed} = \frac{2x}{x\left[\frac{1}{s_1} + \frac{1}{s_2}\right]} = \frac{2}{\frac{1}{s_1} + \frac{1}{s_2}}$$

$$\text{Avg speed} = \frac{2s_1 s_2}{s_1 + s_2} \checkmark$$

→ 3-equal distance interval.

$$\text{Avg speed} = \frac{3}{\frac{1}{s_1} + \frac{1}{s_2} + \frac{1}{s_3}}$$

Question

14

A vehicle travel half the distance with v and remaining half with $2v$ then average speed?

(NEET)

1. $v/3$
2. $4v/3 = 1.33v$
3. $2v/3 = 0.66v$
4. $3v/4$

equal distn interval $= \frac{2 \times v \times 2v}{3v} = \frac{4v}{3}$

MR* concept

$v <$ Avg speed $< 2v$

Question

16

A train has speed 60 km/hr for one hour and 40 km/hr for next half hour, then average speed

1. 50
2. 53.33
3. 48
4. 70

$$\text{Avg speed} = \frac{60 + 40 \times \frac{1}{2}}{(1 + \frac{1}{2})\text{hr}} = \frac{80}{3/2}$$

$$= \frac{160}{3}$$

$$= 53.33$$

Question

17

Object moves for 10s with speed 20 m/s and then it moves with speed 30 m/s for next 10 sec. Then find average speed for complete journey?

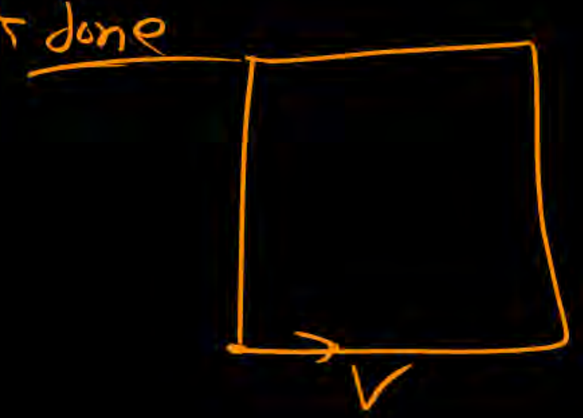
$$A_y = \frac{20+30}{2}$$

$$= 25 \text{ m/s}$$

Question

17

Object is moving with speed ‘S’ on square of side *l* then find average speed and average velocity for given motion



Question

18 copy

Object moves 10m with speed 20 m/s and then it moves with speed 30 m/s for 10 sec. Then find average speed for complete journey?

(a) equal distn ✗

(b) equal time ✗



$$\text{Avg speed} = \frac{\text{Total distance}}{\text{total time}}$$

$$= \frac{10m + 30\times10}{\frac{10}{20} + 10} = \frac{310}{\frac{1}{2} + 10} = \frac{310\times2}{21}$$ m/s

Question

19

Object moves 20m with speed 20 m/s and then it moves with speed 30 m/s for 20m. Then find average speed for complete journey?

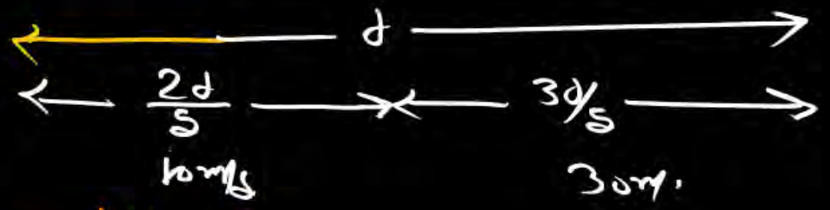
equation

$$\text{Av speed} = \frac{2 \times 20 \times 30}{20 + 30}$$

Question

20

If object moves $\frac{2}{5}^{th}$ distance of journey with speed 10 m/s and remaining with 30 m/s then average speed will be :



$$\text{Avg speed} = \frac{\text{total dist}}{\text{total time}}$$

$$= \frac{d}{\frac{2d}{5\times10} + \frac{3d}{5\times30}} = \frac{1}{\frac{1}{25} + \frac{1}{50}} = \frac{1}{\frac{2+1}{50}} = \frac{50}{3} \text{ m/s}$$

Question

21

A car travels from Kota to Jaipur with speed 30 km/h, and it returns along the same path with speed 60 km/h. Calculate average speed of the car.

equal dist

Question

22 No Copy

A body covers first one-third of the distance with velocity 10 ms^{-1} in same direction, the second one-third with a velocity 20 ms^{-1} and last one-third with a velocity of 30 ms^{-1}. The average velocity of body is

1. 17.8 ms^{-1}
2. 16.4 ms^{-1}
3. 18.3 ms^{-1}
4. 20.2 ms^{-1}



$$\text{Avg Speed} = \frac{3}{\frac{1}{10}+\frac{1}{20}+\frac{1}{30}}$$ ✓

Question 23

Object moves with v_1 for $t/3$ and with v_2 for $2t/3$ then find average speed.

$$\text{Avg speed} = \frac{d_1 + d_1}{t_1 + t_2}$$

$$= \frac{v_1 \frac{t}{3} + v_2 \frac{2t}{3}}{t}$$

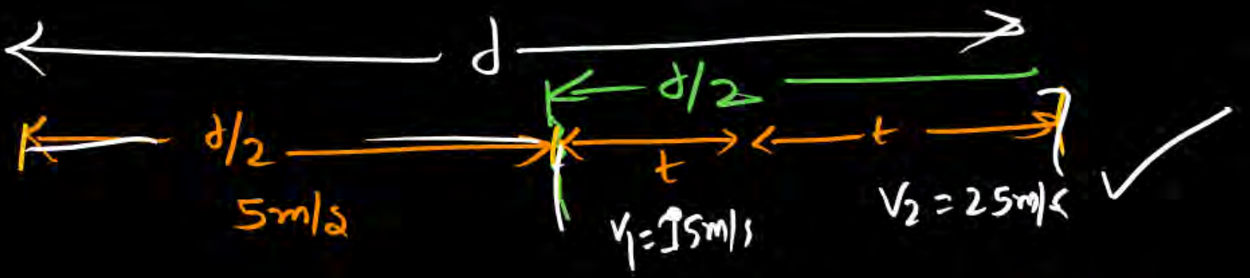
$$= \frac{v_1}{3} + \frac{2v_2}{3}$$

$$\text{Avg speed} = \frac{v_1 + 2v_2}{3}$$ ✓

Question 24

rough Page

A bus travels its half distance of journey with speed 5 m/s. It covers remaining distance in two equal time intervals with speed 15 m/s. Calculate average speed of the bus for the whole journey.



$$V_{Avg\ speed} = \frac{15+25}{2}$$

$$= \frac{40}{2} = 20\ m/s$$

$$Avg\ speed = \frac{2 \times 5 \times 20}{5+20}$$

$$= \frac{2 \times 100}{25} = 8\ m/s$$

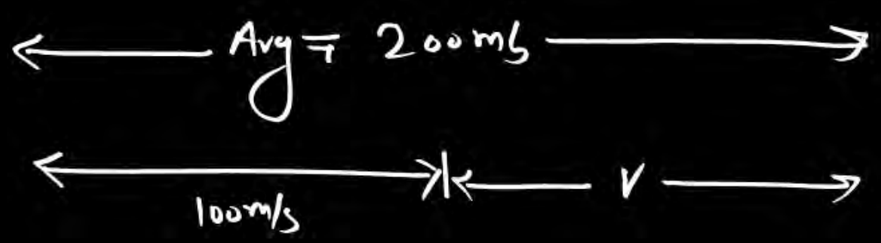
(25) COPY

Avg speed of train b/w two station is 200 m/s if it moves with speed 100 m/s for half distance of total Journey then find speed in 2nd half distance.

VIR SCAM

Soln

Not Possible



$$200 = \frac{2 \times 100 \times V}{100 + V}$$

$100 + V = V$

$100 = 0$

Ø

(a) 300 m/s

(b) 200 m

(c) 100 m/s

(d) 400 m/s

-1 Pen

(e) None of these +1

(26)

Avg speed of train b/w two station is 200 m/s if it moves with speed 400 m/s for half distance of total Journey then find speed in 2nd half distance.

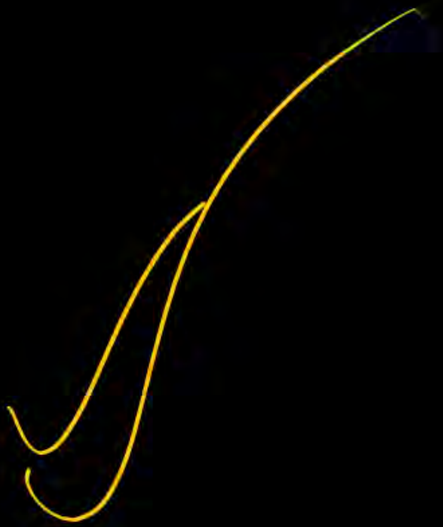
Soln

$$200 = \frac{2 \times 400 \times V}{400 + V}$$

$$4V = 400 + V$$

$$4V - V = 400$$

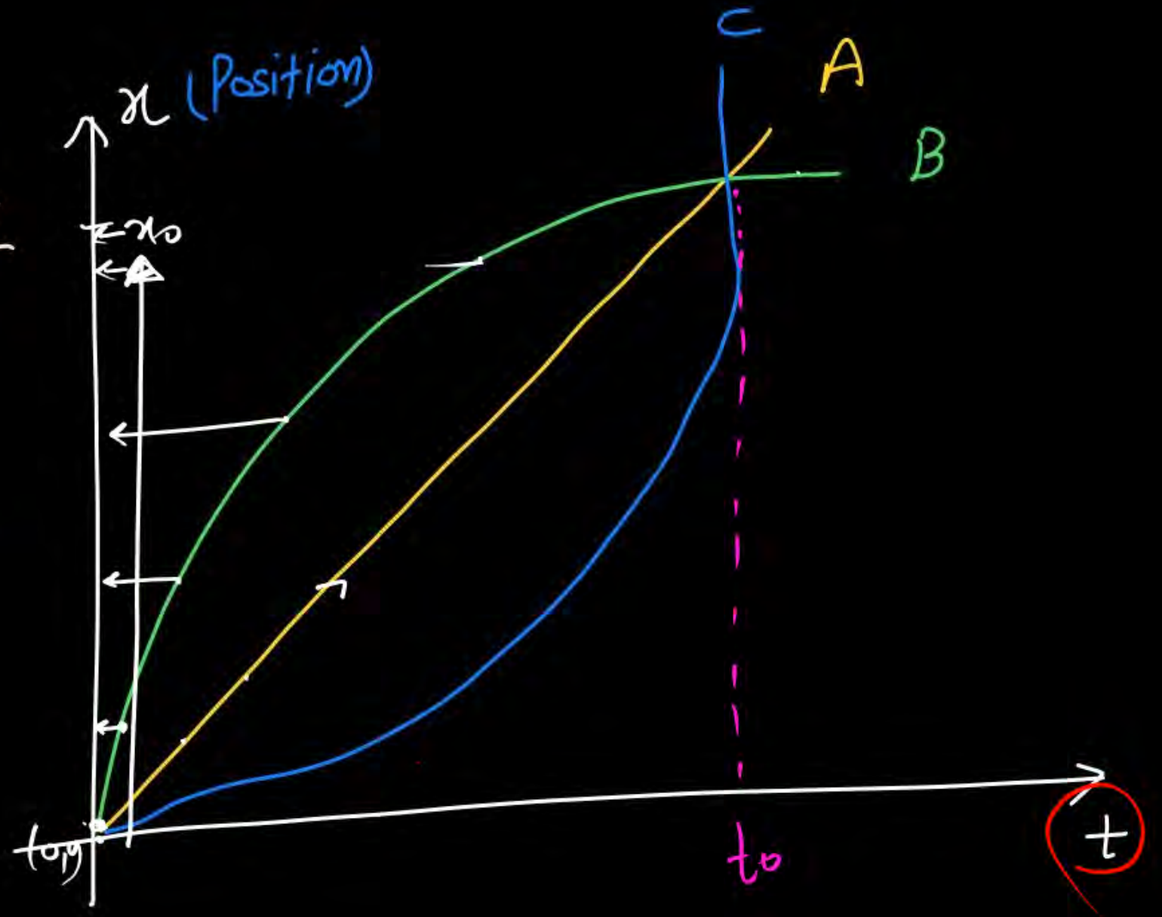
$$V = \frac{400}{3} \text{ m/s}$$



Done

(27)

Position time graph for A, B & C given then compair their Avg. velocity and direction of Avg. velocity in time t_0



Soln Avg velocity same for all A, B, & C

(a) dirⁿ of Avg. velocity same. (57)

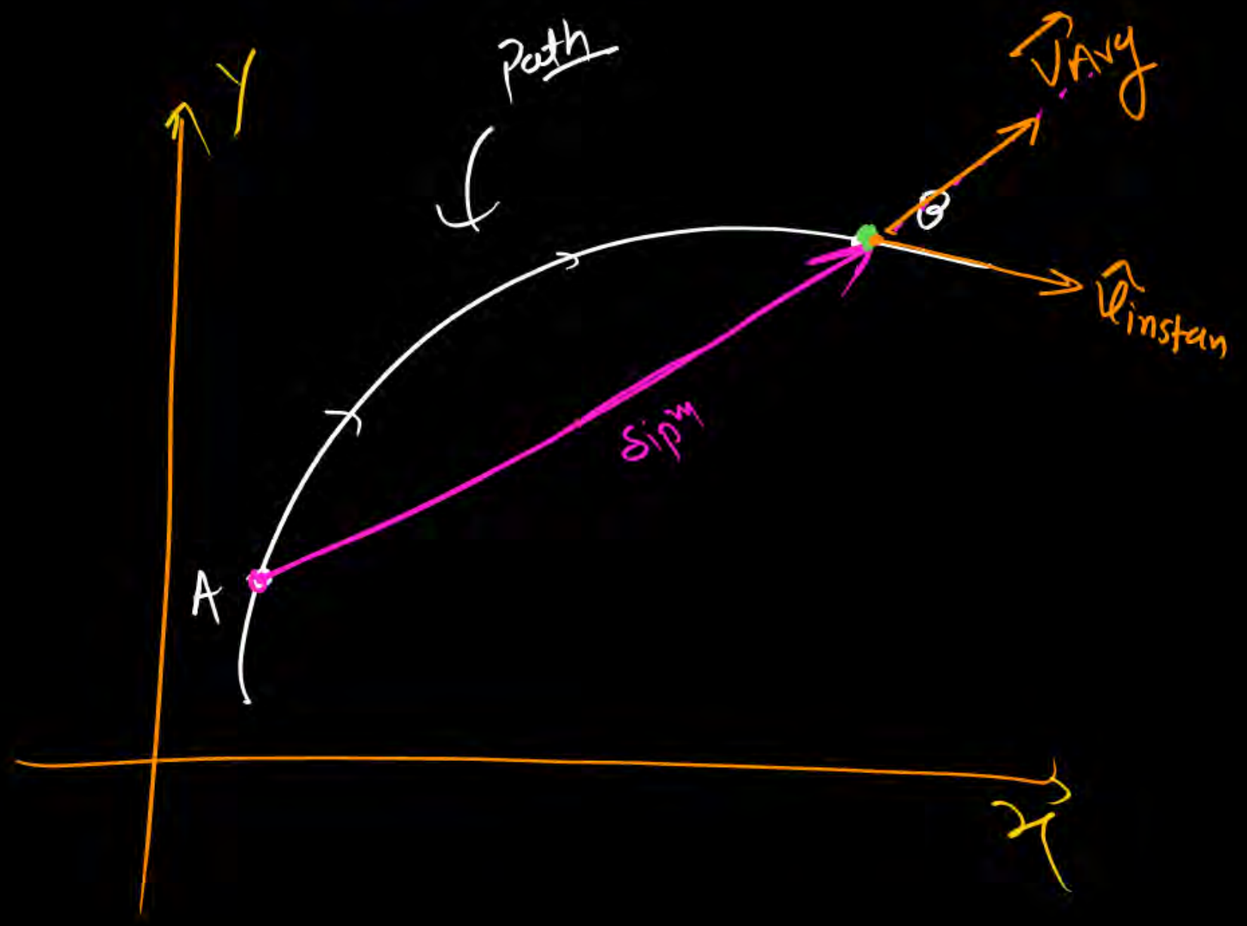
(b) " " " " diffⁿ.

dirⁿ of Avg velocity = along displⁿ

28

dir^{n} of Avg velocity B/w 'A' & 'B'

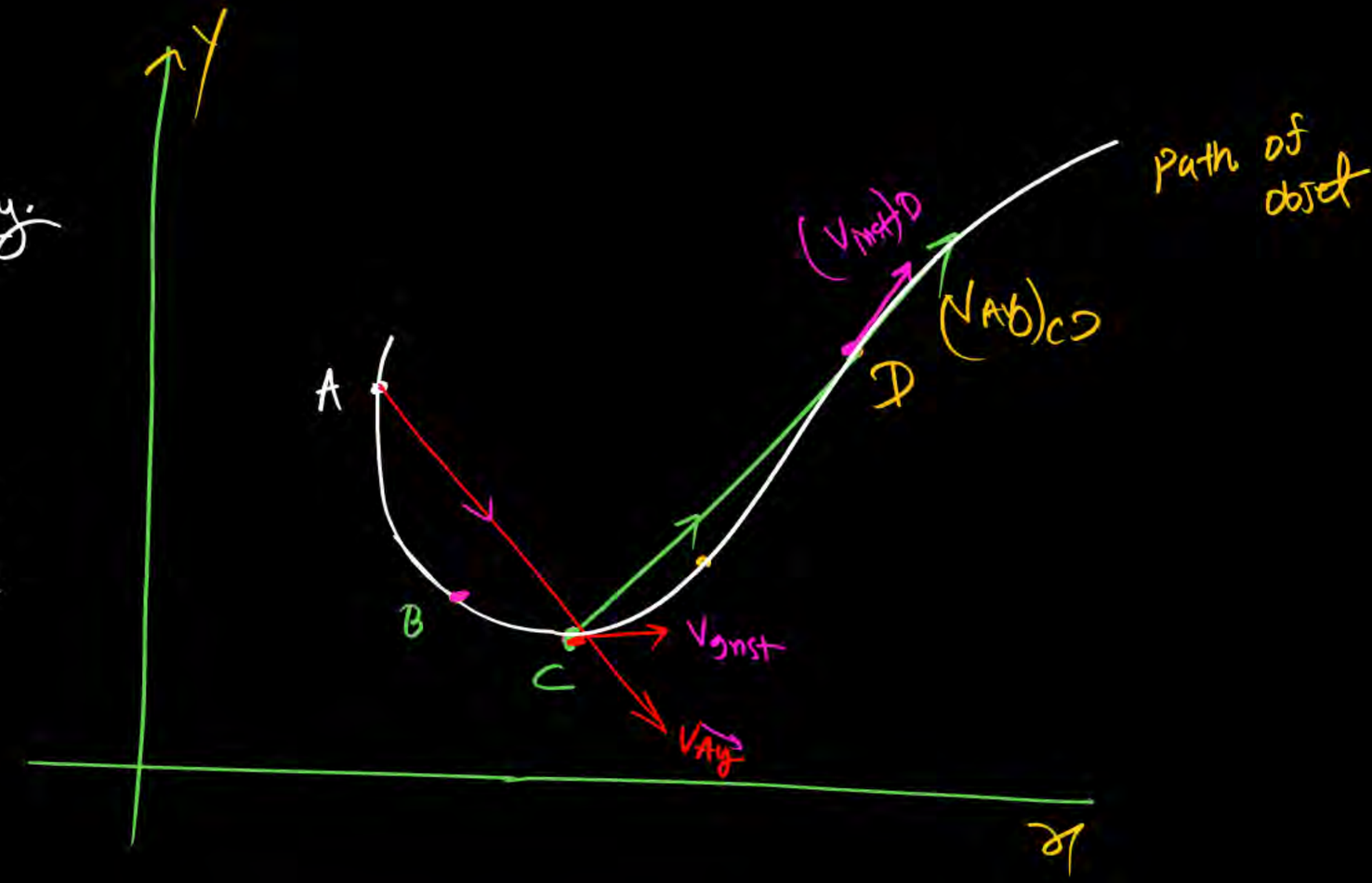
y-x plane.



(29)

① Inst velocity along Particle is Moving along Path

② Avg velocity = along disp^{m} b/w Two Point



Avg velocity $= \frac{\text{Total disp}}{\text{Total}}$	Ans Avg velocity must be zero.	Ans Avg velocity may or may not zero	(32) gf Avg. velocity is zero, then Avg. speed will be ??	(33) gf Avg. velocity is not zero then Avg. speed will be ??
Avg. speed $= \frac{\text{total disp}}{\text{time}}$	(30) gf Avg. speed is zero then Avg velocity ??	(31) gf Avg. speed is not zero. then Avg velocity ??	↳ Ans Avg. speed may or may not zero	↳ Ans Avg. speed must be non-zero

↳ Avg speed $= 0 = \frac{\text{total disp}}{\text{time taken}}$

Instantaneous speed = How fast

- Scalar
- Rate of change in distance w.r.t time
- $S = \frac{dx}{dt}$
  - Slope of distance/time graph

{Magnitude of ~~Inst.~~ Velocity}

Inst. velocity = How fast / where (dirn)

= speed x dirn

- Vector ✓
* Rate of change in displacement ✓

$$\vec{v} = \frac{d\vec{x}}{dt}$$

* Slope of displ^{m}/time graph ✓
* direction → along motion or tangent to path

$|\text{Inst velocity}| = \text{speed}$ → Yes always valid.

Inst. Velocity = Inst. speed × dir^n

MR* Box

use differentiation for gst. speed and gmst. velocity Not for Avg. velocity & Avg. speed.

(34) Diretion of motion changes at instant (time) ??

(a) at A only

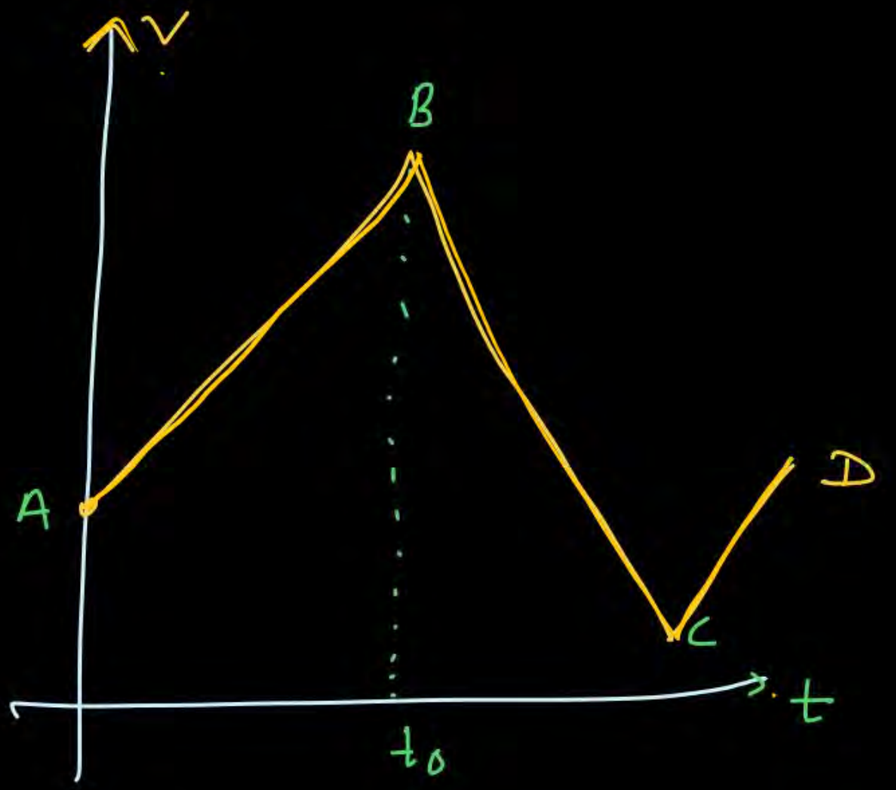
(b) at B only

(c) at C only

(d) at B & C (-1)

(e) Non of these

41% (+1) pen



Save a copy
in notes

(35)

If speed is constant then Avg speed will be ??

(36)

If velocity is constant then Avg velocity will be ??

(37) If object is moving with constant speed then what about velocity ??

(38) If object is moving with constant velocity then what about speed

39 If object moving with variable speed then what about velocity

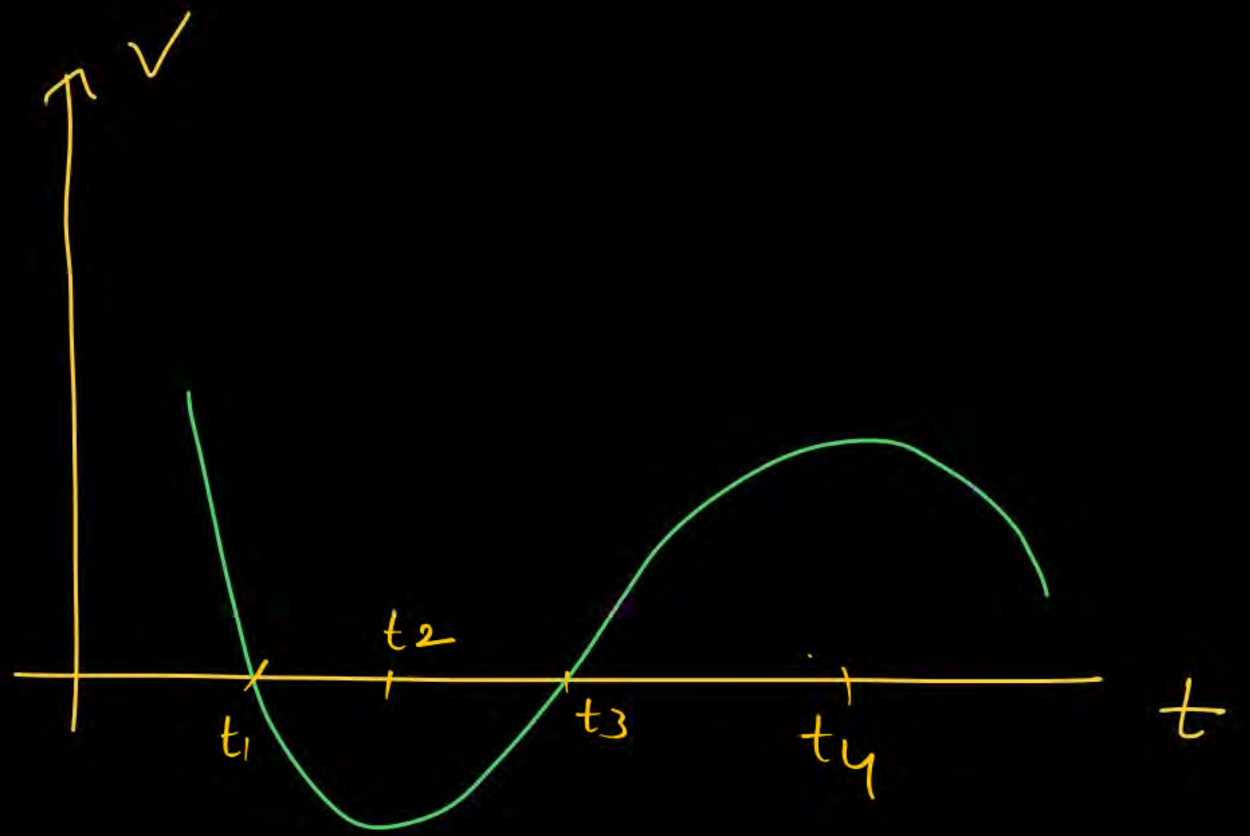
40 If velocity is variable then speed of object will be ??

(41) gf Position $x = \alpha t^2 + \beta t - \gamma$

(i) find itial velocity & position

(ii) find velocit at $t = 2$ sec

(42) object takes U turn at



(43) Position of object $x = t^2 - 4t + 5$ then find distance and displacement in 3se. Avg. speed & Avg. velocity

(45) $x = t^2 - 6t + 4$ find Avg velocity in 4-sec.

(46)

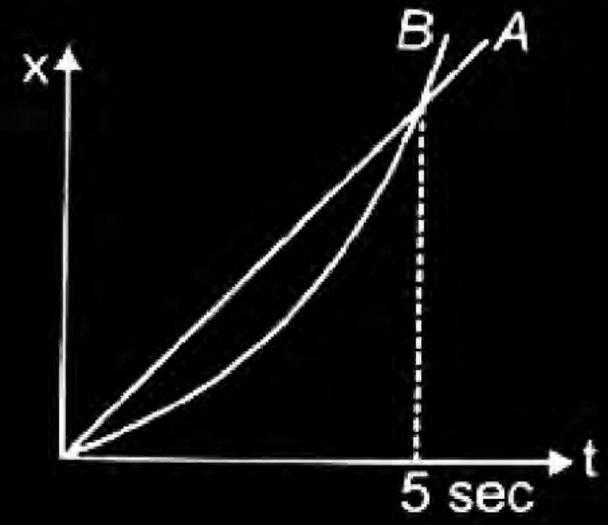
$x = \frac{t^3}{3} - 2t^2 + 4t$ find distance & displacement in 3-sec.

Question

47

Position-time graph for two particles A and B as shown in figure. If both the particles start from origin, and average velocity of A and B during interval of 5 sec are v_A and v_B, then

1. $v_A > v_B$
2. $v_B > v_A$
3. $v_A = v_B$
4. $v_A = 2v_B$

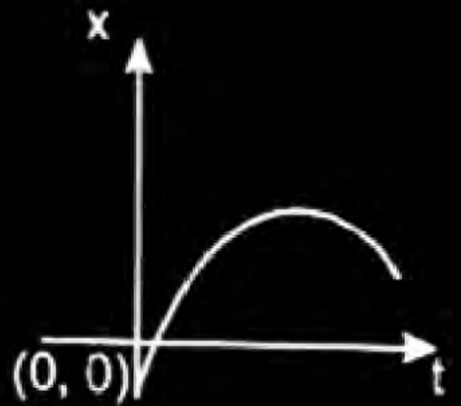


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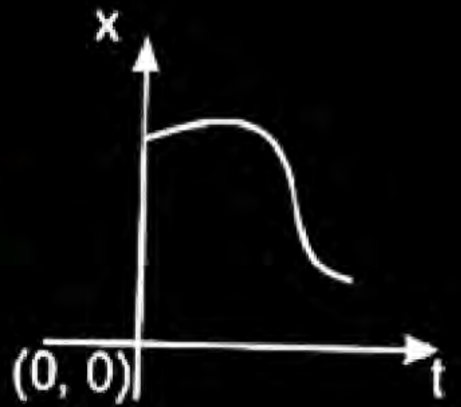
Question

Among the four graphs, there is only one graph for which average velocity = 0 is possible for a particular time interval starting from $t = 0$. Which one is it?

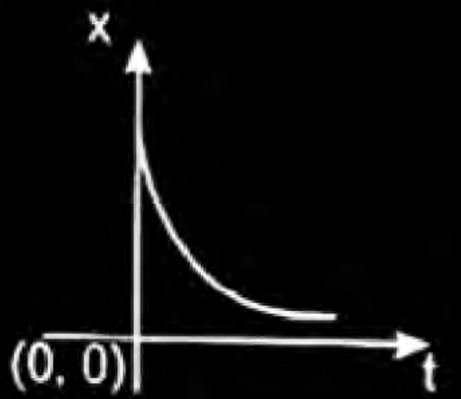
1



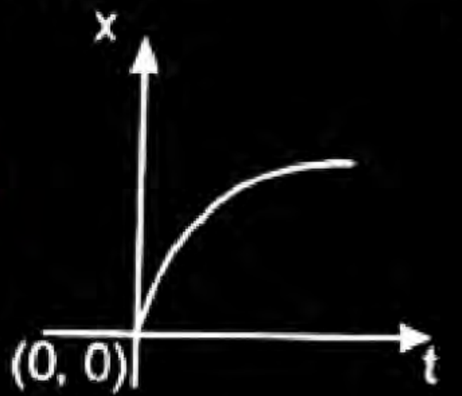
2



3



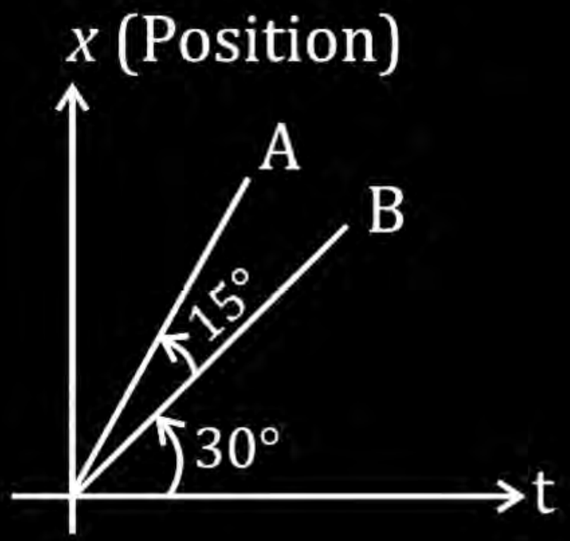
4



Question

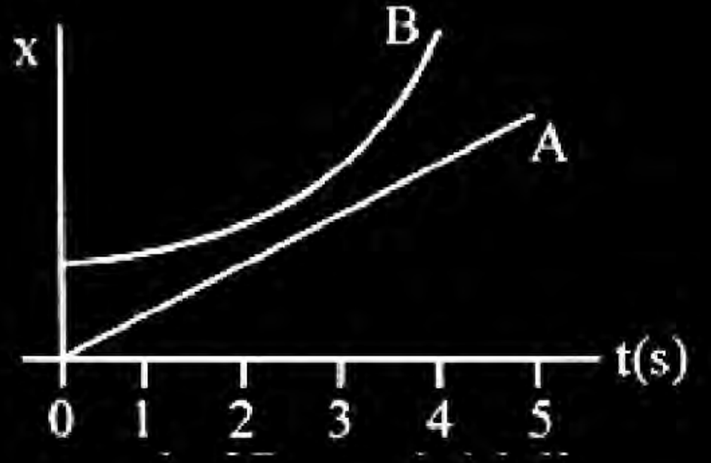
45

Find ratio of velocity of A to B.



Question

In the figure shown below, the position versus time graph of two particles A and B is shown. Select the correct statement:



1. The speed of B was initially greater than that of A and finally less than that of A.
2. The speed of B was initially less than that of A and finally greater than that of A.
3. The speed of B was initially as well as finally greater than that of A.
4. The speed of B was initially as well as finally less than that of A.

Question

51

A particle is moving such that its position coordinates (x, y) are (2m, 3m) at time t = 0, (6m, 7m) at time t = 2s and (13m, 14m) at time t = 5 s. The average velocity vector $\vec{v}_{avg}$ from t = 0 to t = 5 s is:

1. $\frac{1}{5}(13\hat{\imath} + 14\hat{\jmath})$
2. $\frac{7}{3}(\hat{\imath} + \hat{\jmath})$
3. $2\,(\hat{\imath} + \hat{\jmath})$
4. $\frac{11}{5}(\hat{\imath} + \hat{\jmath})$

Question

S2

Object is moving with constant speed then velocity of object:

1. may be variable
2. must be constant
3. must be variable
4. may be zero

Question

53

A particle is executing a circular motion of radius R with a uniform speed v. After completing half the circle, the change in velocity and in speed will be respectively

1. zero, zero
2. $2v$, zero
3. $2v$, $2v$
4. zero, $2v$

Question

54

If object is moving with speed $v = 3t^2$; then find avg. speed in 2-sec.

Thank You